

Exchange Rate Movements and Armenian Inflation: A Regression Analysis with Armenian CPI and FX Data

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Problem Statement and Research Questions

What we study

- Consumer Price Index (CPI) inflation in Armenia: monthly log change of the CPI.
- Exchange rates AMD/USD, AMD/EUR, AMD/RUB: monthly log changes of average official rates.
- Question: do FX movements help explain or predict Armenian inflation?

Research questions

- (Q1) **Contemporaneous association:** Do monthly FX changes help explain monthly CPI inflation beyond lagged inflation and seasonality?
- (Q2) **Short-horizon prediction:** Do current FX changes improve one-month-ahead CPI forecasts?
- (Q3) **Medium-horizon association:** Do twelve-month FX changes help explain twelve-month CPI inflation, even after controlling for persistence?

Data and Transformations

Data sources

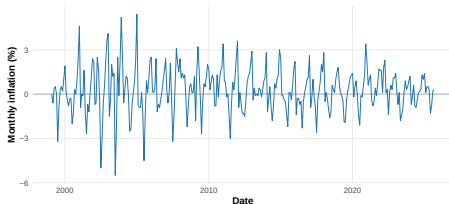
- Monthly CPI (all-items index): ArmStatBank.
- Daily official FX rates vs. AMD (USD, EUR, RUB): Central Bank of Armenia.
- We restrict to the common sample where all series and required lags are available.
- This leaves a balanced sample of $n = 320$ monthly observations of CPI and FX.

Key variables

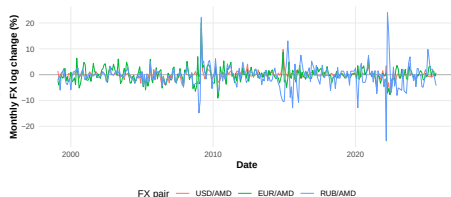
- Monthly inflation: $Y_t = 100(\log C_t - \log C_{t-1})$.
- FX log changes (example USD): $X_t^{\text{USD}} = 100(\log S_t^{\text{USD}} - \log S_{t-1}^{\text{USD}})$, similarly for EUR and RUB. Positive values mean AMD depreciation.
- Twelve-month cumulative variables:

$$Y_t^{(12)} = \sum_{j=0}^{11} Y_{t-j}, \quad X_t^{(12),c} = \sum_{j=0}^{11} X_{t-j}^c, \quad c \in \{\text{USD, EUR, RUB}\}.$$

Time Series Overview



Monthly CPI inflation Y_t

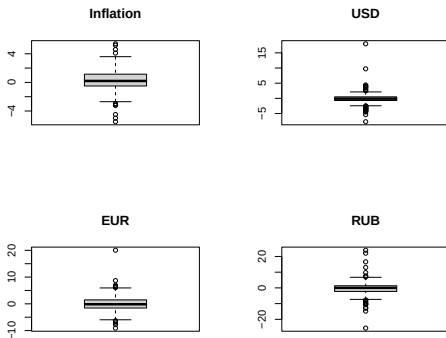


Monthly FX log changes $X_t^{\text{USD}}, X_t^{\text{EUR}}, X_t^{\text{RUB}}$

Inflation and FX series show clear crisis episodes and volatility spikes. This motivates checking for outliers and using diagnostics rather than deleting economically meaningful observations.

Distributional features

- Boxplots confirm heavy tails in both inflation and FX changes.
- Several extreme observations correspond to known crisis periods.
- We keep them in the main regressions and later check robustness using influence diagnostics and trimming.



Boxplots for inflation and FX changes

Seasonality

- Month dummies $D_{m,t}$ capture systematic differences across calendar months.

Q1: Monthly Inflation and Contemporaneous FX

Models

$$(M1) \quad Y_t = \beta_0 + \beta_1 Y_{t-1} + \sum_{m=1}^{11} \gamma_m D_{m,t} + \varepsilon_t,$$

$$(M2) \quad Y_t = \beta_0 + \beta_1 Y_{t-1} + \sum_{m=1}^{11} \gamma_m D_{m,t} + \delta_1 X_t^{\text{USD}} + \delta_2 X_t^{\text{EUR}} + \delta_3 X_t^{\text{RUB}} + \varepsilon_t.$$

Variable	Est.	t	p
Y_{t-1}	0.287	5.19	< 0.001
X_t^{USD}	0.020	0.55	0.58
X_t^{EUR}	0.049	2.10	0.036
X_t^{RUB}	0.001	0.06	0.96

Model comparison

- Adjusted R^2 :

$$(M1) \approx 0.64, \quad (M2) \approx 0.65.$$

- F -test for FX block:

$$F(3, 304) = 3.50, \quad p = \mathbf{0.016}.$$

Interpretation: FX variables are jointly significant at 5%, but their contribution to monthly fit is small.

Q2: Next-Month Inflation Prediction

Models

$$(N1) \quad Y_{t+1} = \alpha_0 + \alpha_1 Y_t + \sum_{m=1}^{11} \eta_m D_{m,t} + u_t,$$

$$(N2) \quad Y_{t+1} = \alpha_0 + \alpha_1 Y_t + \sum_{m=1}^{11} \eta_m D_{m,t} + \theta_1 X_t^{\text{USD}} + \theta_2 X_t^{\text{EUR}} + \theta_3 X_t^{\text{RUB}} + u_t.$$

Variable	Est.	t	p
Y_t	0.260	4.69	< 0.001
X_t^{USD}	0.047	1.32	0.19
X_t^{EUR}	0.012	0.50	0.62
X_t^{RUB}	0.007	0.57	0.57

Model comparison

- Adjusted R^2 :

$$(N1) \approx 0.64, \quad (N2) \approx 0.64.$$

- F -test for FX block:

$$F(3, 303) = 1.79, \quad p = 0.15.$$

Interpretation: current inflation and seasonality drive the forecast; current FX changes do not materially improve one-month-ahead predictions.

Q3: Twelve–Month Inflation and FX

Models

$$(Y1) \quad Y_t^{(12)} = \beta_0^{(12)} + \sum_{m=1}^{11} \gamma_m^{(12)} D_{m,t} + e_t,$$

$$(Y2) \quad Y_t^{(12)} = \beta_0^{(12)} + \sum_{m=1}^{11} \gamma_m^{(12)} D_{m,t} + \delta_1^{(12)} X_t^{(12),\text{USD}} + \delta_2^{(12)} X_t^{(12),\text{EUR}} \\ + \delta_3^{(12)} X_t^{(12),\text{RUB}} + e_t.$$

Variable	Est.	t	p
$X_t^{(12),\text{USD}}$	-0.104	-4.22	< 0.001
$X_t^{(12),\text{EUR}}$	0.027	1.30	0.19
$X_t^{(12),\text{RUB}}$	0.053	3.74	< 0.001

Model comparison

- Adjusted R^2 :

$$(Y1) \approx 0.00, \quad (Y2) \approx 0.07.$$

- F -test for FX block:

$$F(3, 294) = 11.9, \quad p < \mathbf{0.001}.$$

Interpretation: cumulative FX changes are clearly associated with twelve–month inflation, especially AMD/RUB.

Q3 (continued): Controlling for Persistence

Models with lagged twelve-month inflation

$$(Y3) \quad Y_t^{(12)} = \phi_0 + \phi_1 Y_{t-1}^{(12)} + \sum_{m=1}^{11} \kappa_m D_{m,t} + v_t,$$

$$(Y4) \quad Y_t^{(12)} = \phi_0 + \phi_1 Y_{t-1}^{(12)} + \sum_{m=1}^{11} \kappa_m D_{m,t} + \theta_1^{(12)} X_t^{(12),USD} + \theta_2^{(12),EUR} + \theta_3^{(12),RUB} + v_t.$$

Variable	Est.	<i>t</i>	<i>p</i>
$Y_{t-1}^{(12)}$	0.942	47.9	< 0.001
$X_t^{(12),USD}$	0.006	0.64	0.52
$X_t^{(12),EUR}$	0.008	1.16	0.25
$X_t^{(12),RUB}$	0.013	2.79	0.006

Model comparison

- Adjusted R^2 :

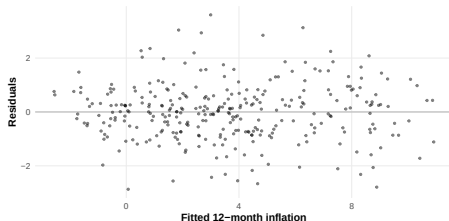
$$(Y3) \approx 0.89, \quad (Y4) \approx 0.89$$

- F -test for FX block:

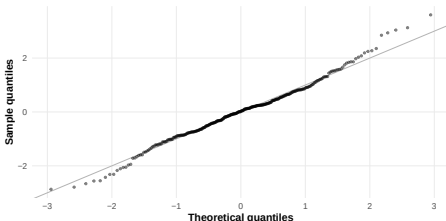
$$F(3, 292) = 5.65, \quad p < \mathbf{0.001}.$$

Interpretation: even after controlling for strong persistence, cumulative AMD/RUB depreciation remains individually and jointly significant.

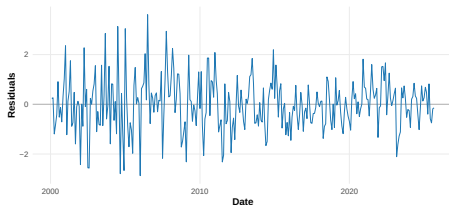
Residual Diagnostics for Twelve-Month Model (Y4)



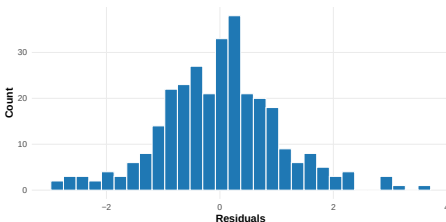
(a) Residuals vs fitted



(c) Normal QQ plot



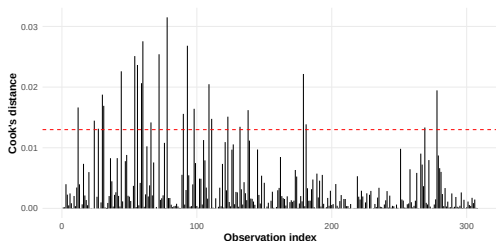
(b) Residuals over time



(d) Residual histogram

- No strong nonlinearity in residuals vs fitted; some heteroskedasticity at extremes.

Influence Diagnostics and Robustness



Cook's distance for model (Y4)

Influence

- Horizontal reference: $4/n \approx 0.01$ as a rough influence threshold.
- Only a handful of observations exceed this level; they are months with very large FX and inflation moves in crisis periods.

Robustness

- Re-estimating (Y4) after trimming the most extreme FX changes leaves the sign and significance of the AMD/RUB coefficient essentially unchanged.
- Together with residual diagnostics, this supports the linear model as a reasonable approximation for average relationships between inflation and FX in our sample.

Summary by question

- **Q1 (monthly association).** Monthly FX changes are jointly significant in explaining monthly inflation, but the effect is small and mainly driven by AMD/EUR; adjusted R^2 hardly moves.
- **Q2 (one-month-ahead prediction).** After controlling for current inflation and seasonality, current FX changes are neither individually nor jointly significant for short-horizon forecasts.
- **Q3 (twelve-month horizon).** Cumulative FX changes matter: AMD/RUB shows a robust positive association with year-on-year inflation, even after controlling for strong persistence.

Interpretation

- In our deliberately simple linear framework, short-run inflation dynamics are dominated by persistence and seasonality.
- Exchange rate movements, especially against the ruble, are more visible at a twelve-month horizon than month by month.

Model adequacy

- Residual and influence diagnostics for the twelve-month model (Y4) do not reveal severe violations of the linear model assumptions.
- Together with robustness checks, this suggests that, despite its simplifications, the linear specification is adequate for summarising average relationships between Armenian CPI inflation and FX movements in our sample.

Thank you for your attention!

Questions are welcome.